TurboWin (version 3.02) Quality Control

Level-1 checks: input form checks (errors/not possible to insert)
Level-1b checks: mandatory parameters/groups (errors)
Level-2 checks: consistency all inserted parameters (errors)
Level-3 checks: warnings

Level-1 checks

Call sign

Parameter(s)	**Not possible**
Call sign	Outside range: 0-9, a-z, A-Z (per character)

Date + Time

Parameter(s)	**Error**
Year	Outside range: 2001-2010
Month	Outside range: January–December or outside range: 1–12
Day	Outside range 1–31
Time	Outside range 0–23
Year – month – day	Year = 2004, 2008, 2012, 2016 + Month = 2 (or February) + Day > 29
Year – month – day	Year not 2004, 2008, 2012, 2016 + Month = 2 (or February) + Day > 28
Month – day	Month = 4,6,9,11 + Day > 30

Position, Course, Speed

Parameter(s)	**Error**
Latitude degrees	Not present
Latitude minutes	Outside range : 00-59 min
Latitude hemisphere	Not South or North
Latitude	Outside range: 0.0–90.0º
Long degrees	Not present
Long tenths	Outside range: 00-59 minutes
Long hemisphere	Not West or East
Long	Outside range: 0.0–180.0º
Course	Outside range: 1–360º, "stationary", unknown
Speed	Outside range: 0–'>40' knots, unknown
Course – speed	Course = "stationary" + Speed not 0
Speed – course	Speed = 0 + course not "stationary"

Wind

	Parameter(s)	**Error**
	Speed	Outside range: 0–200, blank
	Direction -source	Source = apparent BOW + dir. outside range: 0–360º, "variable", "calm", blank
	Direction -source	Source = not apparent BOW + dir. outside range: 1–360º, "variable", "calm", blank
	Speed – units	speed = present + units not present
	Speed – source	Speed = present + source not present
	Direction – source	Direction present + source not present
	Direction – speed	Direction = "variable" + speed outside range 0–2 m/s
	Direction – speed	Direction = "variable" + speed outside range 0–4 knots
	Source – speed – direction	Source = apparent + speed or direction not present
2.1	VOS-Clim - Source – speed – direction	VOS-Clim + source = estimated + speed or direction not present
	Source – ground course	Source = apparent + course outside range 1-360º, "stationary"
	Source – ground speed	Source = apparent + ground speed outside range 0-50 knots

	Source – ground speed tenths	Source = apparent + ground speed tenths outside range 0-9
	Source – heading	Source = apparent BOW + heading outside range 1-360º, blank
	Source – ground course – ground speed	Source = apparent + ground course = "stationary" + ground speed not 0.0
	Source – ground speed – ground course	Source = apparent + ground speed = 0.0 + ground course not "stationary"
	Station – settings	Station = fixed + anemometer settings not available
2.1	VOS-Clim – add. data	VOS-Clim + diff. summer line – water line outside range –99 – 99 metres
2.1	VOS-Clim – add. data	VOS-Clim + Height deck cargo outside range 0 – 99 metres
2.1	VOS-Clim – wind speed – summer line	VOS-Clim + wind speed = present + diff. summer max l.l. – water line not present
2.1	VOS-Clim – wind speed – height deck cargo	VOS-Clim + wind speed = present + height deck cargo not present
	Parameter(s)	**Not possible**
2.1	VOS-Clim – add. data	Outside range: 0-9, "-" (per character)
2.1	VOS-Clim – add. data	Outside range: 0-9 (per character)

Waves

	Parameter(s)	**Error**
	Method – period	Method = measured + period outside range: 0-50 sec, blank
	Method – height metres	Method = measured + height metres outside range: 0-50 metres, blank
	Method – height tenths	Method = measured + height tenths outside range: 0-9, blank
	Method – height metres – height tenths	Method = measured + height metres present + height tenths not present
	Method – height tenths – height metres	Method = measured + height tenths present + height metres not present
	Method – sea height	Method = estimated + sea height outside range: 0 – 49.0 metres, "confused", blank
	Method – sea period	Method = estimated + sea period outside range: 0 – 50 sec, "confused", blank
	Method – 1st swell height	Method = estimated + 1st swell height outside range: 0 – 49.0 metres, "confused", blank
	Method – 1st swell period	Method = estimated + 1st swell period outside range: 0 – 50 sec, "confused", blank
	Method – 1st swell direction	Method = estimated + 1st swell dir outside range: 0 – 360 degr, "confused", "no swell", blank
	Method – 2nd swell height	Method = estimated + 2nd swell height outside range: 0 – 49.0 metres, "confused", blank
	Method – 2nd swell period	Method = estimated + 2nd swell period outside range: 0 – 50 sec, "confused", blank
	Method – 2nd swell direction	Method = estimated + 2nd swell dir outside range: 0 – 360 degr, "confused", "no swell", blank
	Method – sea period – sea height	Method = estimated + sea period = 0 sec + sea height > 2.2 metres
	Method – sea height – sea period	Method = estimated + sea height = 0 metres + sea period > 5 sec
	Method – sea height – sea period	Method = estimated + sea height = 0 metres + sea period = "confused"
	Method – 1st swell period –1st swell height	Method = estimated + 1st swell period = 0 sec + 1st swell height > 2.2 metres
	Method – 1st swell height – 1st swell period	Method = estimated + 1st swell height = 0 metres + 1st swell period > 5 sec
	Method – 1st swell dir – 1st swell height	Method = estimated + 1st swell dir = "no swell" (or 0) + 1st swell height not 0, blank
	Method – 1st swell dir – 1st swell period	Method = estimated + 1st swell dir = "no swell" (or 0) + 1st swell period not 0, blank
	Method – 1st swell height – 1st swell dir	Method = estimated = 1st swell height = 0 metres + 1st swell dir = "confused"

	Method – 2nd swell period –2nd swell height	Method = estimated + 2nd swell period = 0 sec + 2nd swell height > 2.2 metres
	Method – 2nd swell height – 2nd swell period	Method = estimated + 2nd swell height = 0 metres + 2nd swell period > 5 sec
	Method – 2nd swell dir – 2nd swell height	Method = estimated + 2nd swell dir = "no swell" (or 0) + 2nd swell height not 0, blank
	Method – 2nd swell dir – 2nd swell period	Method = estimated + 2nd swell dir = "no swell" (or 0) + 2nd swell period not 0, blank
	Method – 2nd swell height – 2nd swell dir	Method = estimated + 2nd swell height = 0 metres + 2nd swell dir = "confused"
	Method = 1st swell dir – 2nd swell dir	Method = estimated + 1st swell dir = blank + 2nd swell dir not blank
	Method = 1st swell dir – 2nd swell dir	Method = estimated + 1st swell dir = "confused" + 2nd swell dir not "confused",blank
	Method = 1st swell dir – 2nd swell dir	Method = estimated + 1st swell dir = "no swell" (or 0) + 2nd swell dir not "no swell" (or 0), blank
3.0	~~Method = 2nd swell dir – 1st swell dir~~	~~Method = estimated + 2nd swell dir = "no swell" (or 0) + 1st swell dir not "no swell" (or 0), blank~~
	Method – 1st swell – 2nd swell	Method = estimated + 1st swell dir = 2nd swell dir + 1st swell period = 2nd swell period + 1st swell height = 2nd swell height
	Parameter(s)	**Not possible**
3.0	Message form - waves	SUS, SUP, AUX + sea (period, height)
3.0	Message form - waves	SUS, SUP, AUX + swell (direction, period, height; both systems)

Pressure

Parameter(s)	**Error**
Reading indication	Not present
Barometer height	Not present
Pressure hPa – press. tenths*	pressure hPa present + pressure tenths not present
Reading tenths – press. hPa*	Pressure tenths present + pressure hPa not present
Pressure	Outside range: 890.0 – 1060.0 hPa, blank
Amount tendency hPa – amount tendency tenths**	Amount tendency hPa present + amount tendency tenths not present
Amount tendency tenths – amount tendency hPa**	Amount tendency tenths present + amount tendency hPa not present
Amount tendency**	Outside range 0.0 – 50.0 hPa, blank

* AUX excepted; **SUS, SUP, AUX excepted

	Parameter(s)	**Not possible**
	Characteristic tendency	Outside range: 0 – 8, blank (not determined)
3.0	Message form – amount tend.	SUS, SUP, AUX + amount tendency
3.0	Message form – char. tend.	SUS, SUP, AUX + characteristic tendency
3.0	Message form – pressure tenths	AUX + pressure tenths

Temperatures

	Parameter(s)	**Error**
	Air temp degr – air temp tenths*	Air temp degr present + air temp tenths not present
	Air temp tenths – air temp degr*	Air temp tenths present + air temp tenths not present
	Air temp	Outside range –50.0 - +60.0 °C, blank
	Wet-bulb degr – wet-bulb tenths**	Wet-bulb degr present + wet-bulb tenths not present
	Wet-bulb tenths – wet-bulb degr**	Wet-bulb tenths present + wet-bulb tenths not present
	Wet-bulb - State wet-bulb**	Wet-bulb present + state wet-bulb "not determined"
3.0	Wet-bulb - State wet-bulb**	Wet bulb > 0.0 °C + state Wet-bulb frozen
	Wet-bulb**	Outside range –50.0 - +50.0 °C, blank
	Sst degr – sst tenths***	Sst degr present + sst tenths not present
	Sst tenths – sst degr***	Sst tenths present + sst degr not present

	Sst - sst exposure***	Sst present + sst exposure = “not determined”
	Sst***	Outside range –2.0 - +50.0 °C, blank
	Relative humidity**	Rel. hum outside range 1 – 100 %, blank
	Air temp – wet-bulb**	Wet-bulb > air temp

*AUX excepted; **SUS, SUP, AUX excepted; ***SUP, AUX excepted

	Parameter(s)	**Not possible**
3.0	Message form – air temp	AUX + air temp tenths
3.0	Message form - wet-bulb	SUS, SUP, AUX + wet bulb degrees
3.0	Message form - wet-bulb	SUS, SUP, AUX + wet bulb tenths
3.0	Message form - rel.hum.	SUS, SUP, AUX + relative humidity
3.0	Message form – sst	SUP, AUX + sst degrees
3.0	Message form – sst	SUP, AUX + sst tenths
3.0	Message form – sst	SUP, AUX + sst exposure

Present weather

Parameter(s)	**Not possible**
Present weather (ww*)	Outside range: 00-99*, not determined(blank)

*WMO FM 13-XI SHIP code; however observer selects plain text

Past weather

Parameter(s)	**Not possible**
1st past weather (W1*)	Outside range: 0–9*, blank
2nd past weather (W2*)	Outside range: 0–9*, blank
1st – 2nd past weather	2nd past weather >= 1st past weather (W2 > W1: error if classic form)

*WMO FM 13-XI SHIP code; however observer selects plain text

Parameter(s)	**Error**
1st – 2nd past weather	1st past weather = 2* + 2nd past weather = 1*
1st – 2nd past weather	1st past weather = 2* + 2nd past weather = 0*
1st – 2nd past weather	1st past weather = 1* + 2nd past weather = 0*

*WMO FM 13-XI SHIP code; however observer selects plain text

Visibility

Parameter(s)	**Not possible**
Visibility (VV*)	Outside range: 90-99*, not determined (blank)

*WMO FM 13-XI SHIP code; however observer selects plain text

Clouds Low (Cl)

Parameter(s)	**Not possible**
Cl*	Outside range 0-9*, not determined (blank)

*WMO FM 13-XI SHIP code; however observer selects photo/code figure

	Parameter(s)	**Not possible**
3.0	Message form -Cl	AUX + Cl

Clouds Middle (Cm)

Parameter(s)	**Not possible**
Cm*	Outside range 0-9*, not determined (blank)

*WMO FM 13-XI SHIP code; however observer selects photo/code figure

	Parameter(s)	**Not possible**
3.0	Message form -Cm	AUX + Cm

Clouds High (Ch)

Parameter(s)	**Not possible**
Ch*	Outside range 0-9*, not determined (blank)

*WMO FM 13-XI SHIP code; however observer selects photo/code figure

	Parameter(s)	**Not possible**

3.0	Message form -Ch	AUX + Ch

Cloud cover + height

Parameter(s)	Not possible
Total cover (N*)	Outside range: 0-9*, “not determined”
Low cover (Nh*)	Outside range: 0-9*, “not determined”
Height (h*)	Outside range: 0-9*, “not determined”

*WMO FM 13-XI SHIP code; however observer selects plain text

	Parameter(s)	Error
	Total cover – low cover*	Total cover = “cloudless” + low cover not “0/8”
	Total cover – height*	Total cover = “cloudless” + height not “cloudless”
	Height – total cover*	Height = “cloudless” + total cover not “cloudless” (not in classic form)
3.0	~~Height – total cover~~	~~Height = “0-50m” + total cover = “cloudless”~~
	Total cover – height*	Total cover = “obscured” + height not “not determined”
	Total cover – low cover*	Total cover = “obscured” + low cover not “obscured”, “not determined”
	Total cover – height*	Total cover = “not determined” + height not “not determined”
	Total cover – low cover*	Total cover = “not determined” + low cover not “not determined”
	Total cover – low cover*	Total cover < low cover

*AUX excepted

	Parameter(s)	Not possible
3.0	Message form – low cover	AUX + low cover
3.0	Message form – heght	AUX + height

Icing

Parameter(s)	Not possible
Cause (Is*)	Outside range: 1-5*, “not determined”
Rate (Rs)	Outside range: 0-4*, “not determined”

*WMO FM 13-XI SHIP code; however observer selects plain text

Parameter(s)	Error
Thickness (EsEs)	Outside range: 0-99 centimetres, blank

Ice

Parameter(s)	Not possible
Development (Si*)	Outside range: 0-9*, “unable to report”, blank
Land ice (bi*)	Outside range: 0-9*, “unable to report”, blank
Ice edge (Di)	Outside range: 0-9*, “unable to report”, blank
Sea ice (ci)	Outside range: 0-9*, “unable to report”, “not determined”, blank
Ice situation + trend (zi)	Outside range: 0-9*, “not determined”, blank

*WMO FM 13-XI SHIP code; however observer selects plain text

Observer

Parameter(s)	error
Mode – surname	Mode = add + surname not present
Mode – initials	Mode = add + initials not present
Mode – rank	Mode = add + rank not present

Captain

Parameter(s)	error
Mode – surname	Mode = add + surname not present
Mode – initials	Mode = add + initials not present
Mode – joining	Mode = add + joining not present

National groups

Parameter(s)	Not possible
1st group	Outside range: 0-9 (per character)

1st group	Number of characters > 5
2nd group	Outside range: 0-9 (per character)
2nd group	Number of characters > 5
3td group	Outside range: 0-9 (per character)
3td group	Number of characters > 5
4th group	Outside range: 0-9 (per character)
4th group	Number of characters > 5
5th group	Outside range: 0-9 (per character)
5th group	Number of characters > 5

Parameter(s)	**Error**
1st group	Number of characters not 5 (blank = no error)
2nd group	Number of characters not 5 (blank = no error)
3td group	Number of characters not 5 (blank = no error)
4th group	Number of characters not 5 (blank = no error)
5th group	Number of characters not 5 (blank = no error)

Level-1b checks

Date/Time

Parameter(s)	error
Mode – Date/time	Mode = obs completed _+ date/time not present

Position

Parameter(s)	error
Mode – position	Mode = obs completed + position not present

Call sign

Parameter(s)	error
Mode – call sign	Mode = obs completed + call sign not present

Level-2 checks

Wind - waves

	Parameter(s)	error
	Wind speed - sea period	Wind speed = 0 and sea period not 0, blank
	Wind speed - sea height	Wind speed = 0 and sea height not 0, blank
	Wind speed - sea height	Wind speed in range 0-3 m/s + sea height > 9.7 metres
	Wind speed - sea height	Wind speed in range 0-6 knots + sea height > 9.7 metres
2.1	~~Wind speed - sea height~~	~~Wind speed > 5 m/s + sea height = 0 metres~~
2.1	~~Wind speed - sea height~~	~~Wind speed > 10 knots + sea height = 0 metres~~
2.1	~~Wind speed - sea height~~	~~Wind speed > 21 m/s + sea height < 2.3 metres~~
2.1	~~Wind speed - sea height~~	~~Wind speed > 41 knots + sea height < 2.3 metres~~

Air pressure

Parameter(s)	Error
Characteristic tendency - amount tendency	Characteristic = the same-steady (a = 4) + amount tendency (ppp) not 0

Cloud cover – cloud types

Parameter(s)	error
Total cloud cover - Clouds low (Cl)*	Total cloud cover = “cloudless” + Cl not “no clouds Cl”
Total cloud cover - Clouds middle (Cm)*	Total cloud cover = “cloudless” + Cm not “no clouds Cm”
Total cloud cover - Clouds high (Ch)*	Total cloud cover = “cloudless” + Ch not “no clouds Ch”
Cloud types (Cl,Cm,Ch) - total cloud cover*	Cl+Cm+Ch = “no clouds” + total cloud cover not “cloudless”
Low cloud cover – Cl*	Low cloud cover = 0/8 + Cl not “no clouds Cl”
Low cloud cover - Cl – Cm*	Low cloud cover = 0/8 + Cl = “no clouds Cl” + Cm not “no clouds Cm”
Cl + Cm + low cloud cover*	Cl = “no clouds Cl” + Cm = “no clouds Cm” + low cloud cover not 0/8
Low cloud cover – Ch*	Low cloud cover = 8/8 + Ch in range 1-9,
Low cloud cover - Cl –Ch*	Low cloud cover = 8/8 + Cl in range 1-9 + Cm in range 1-9

*AUX excepted

Cloud cover – present weather

Parameter(s)	error
Present weather – total cloud cover	Present weather (ww) = 43,45,47,49* + total cloud cover not “obscured”, “not determined”
Present weather – Cl	Present weather (ww) = 43,45,47,49* + Cl not “not determined”
Present weather – Cm	Present weather (ww) = 43,45,47,49* + Cm not “not determined”
Present weather – Ch	Present weather (ww) = 43,45,47,49* + Ch not “not determined”
Total cloud cover – present weather	Total cloud cover = “cloudless” + present weather in range 50-69*

*WMO FM 13-XI SHIP code; however observer selects plain text

Cloud type – cloud height

Parameter(s)	Error
Cl – cloud height*	Cl in range 1-9 + cloud height >= 2500 metres (8000 ft), “cloudless”
Cl – Cm – Ch - cloud height*	Cl = ”no clouds Cl” + Cm = “no clouds Cm” + Ch in range 1-9 + cloud height < 200 metres (600 ft)

*AUX excepted

Visibility – present weather

Parameter(s)	Error
Present weather - visibility	Present weather in range 42-49* + visibility in range 95-99*

Present weather - visibility	Present weather = 40* + visibility in range 90-93*

*WMO FM 13-XI SHIP code; however observer selects plain text

Present weather – air-temperature

	Parameter(s)	Error
	Present weather – temp	Temperature > 20.0 ºC + present weather in range 36-39*
	Present weather – temp	Temperature > 20.0 ºC + present weather in range 48-49*
	Present weather – temp	Temperature > 20.0 ºC + present weather in range 56-57*
	Present weather – temp	Temperature > 20.0 ºC + present weather in range 66-67*
	Present weather – temp	Temperature > 20.0 ºC + present weather in range 68-69*
	Present weather – temp	Temperature > 20.0 ºC + present weather in range 70-75*
	Present weather – temp	Temperature > 20.0 ºC + present weather in range 83-86*
2.1	~~Present weather – temp~~	~~Temperature > 0.0 ºC + present weather in range 48-49*~~
2.1	~~Present weather – temp~~	~~Temperature > 2.0 ºC + present weather in range 56-57*~~
2.1	~~Present weather – temp~~	~~Temperature > 2.0 ºC + present weather in range 66-67*~~
2.1	~~Present weather – temp~~	~~Temperature > 2.0 ºC + present weather in range 76-79*~~
2.1	Present weather - temp	Temperature > 20.0 ºC + present weather in range 76-79*

*WMO FM 13-XI SHIP code; however observer selects plain text

Icing – air-temperature

Parameter(s)	Error
Temperature – type icing (Is*)	Temperature > 20.0 ºC + type of icing not blank
Temperature – thickness (EsEs*)	Temperature > 20.0 ºC + ice thickness not blank
Temperature – rate icing (Rs*)	Temperature > 20.0 ºC + rate of ice accretion not blank

*WMO FM 13-XI SHIP code; however observer selects plain text

Ice – air-temperature

Parameter(s)	Error
Temperature – sea ice (ci*)	Temperature > 25.0 ºC + sea ice not blank
Temperature – development (Si*)	Temperature > 25.0 ºC + development nor blank
Temperature – land ice (bi*)	Temperature > 25.0 ºC + land ice not blank
Temperature – ice edge (Di*)	Temperature > 25.0 ºC + ice edge not blank
Temperature – ice situation + trend (zi*)	Temperature > 25.0 ºC + ice situation + trend not blank

*WMO FM 13-XI SHIP code; however observer selects plain text

Level-3 checks

All

	Parameter(s)	Warning (confirmation required)
	System timer - year + month + day + time	Time difference between inserting date + time and sending obs > 0.5 hour
	Position – land-sea mask	Possible land position
	Speed ship (Vs)	> 35 knots
	Wind speed [m/s]	> 28 m/s
	Wind speed [knots]	> 55 knots
	Wind waves period	> 25 seconds
	Wind wave height	> 12.2 metres
	First swell period	> 25 seconds
	First swell height	> 12.2 metres
	Second swell period	> 25 seconds
	Second swell height	> 12.2 metres
2.1	Wind speed [m/s]- sea height	Wind speed > 5 m/s + sea height = 0 metres
2.1	Wind speed [kts]- sea height	Wind speed > 10 knots + sea height = 0 metres
2.1	Wind speed [m/s]- sea height	Wind speed > 21 m/s + sea height <= 2.2 metres
2.1	Wind speed [kts] - sea height	Wind speed > 41 knots + sea height <= 2.2 metres
	Air temp	> 50.0 ºC
	Air temp	< -20.0 ºC
	Air temp – type icing (Is*)	Air temp > 4.0 ºC + type of icing not blank
	Air temp – thickness (EsEs*)	Air temp > 4.0 ºC + ice thickness not blank
	Air temp – rate icing (Rs*)	Air temp > 4.0 ºC + rate icing not blank
	Air pressure	< 910.0 hPa
	Air pressure	> 1050.0 hPa
	Amount pressure tendency (ppp*)	> 30.0 hPa
	Amount pressure tendency (ppp*) – characteristic (a*)	Amount pressure tendency present + characteristic not present
	Sea surface temp	> 35.0 ºC
	Thickness ice accretion (EsEs*)	> 20 centimetres
	Present weater (ww*) – icing	Ww=48,49 + type icing (Is*) + thickness (EsEs*) + rate icing (Rs*) = blank
	Present weater (ww*) – icing	Ww=56,57 + type icing (Is*) + thickness (EsEs*) + rate icing (Rs*) = blank
	Present weater (ww*) – icing	Ww=66,67 + type icing (Is*) + thickness (EsEs*) + rate icing (Rs*) = blank
	Observer	If not selected

*WMO FM 13-XI SHIP code; however observer selects plain text

Miscellaneous

Parameter(s)	
System timer - all parameters	If time difference between sending obs + present system time > 1.0 hour then all inserted parameters are set invalid (but still present; must be confirmed again)

<u>Abbreviations</u>
SUS = supplementary ship's code **with** sst (sea surface temperature) instruments applied
SUP = supplementary ship's code **without** sst (sea surface temperature) instruments applied
AUX = auxiliary ship's code
VOS-Clim = Voluntary Observing Ship Climate Project
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